A Preliminary Mini Project Report On

"Private SaaS Cloud using HDFS"

Submitted to

Savitribai Phule Pune University

In partial fulfillment of the requirement for the award of

T.E Computer Engineering By

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UNDER THE GUIDANCE OF

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DEPARTMENT OF COMPUTER ENGINEERING

Dhole Patil College of Engineering Pune

AY 2025-2026

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AY 2025-2026

CERTIFICATE

This is to certify that the project entitled

Private SaaS Cloud using HDFS

Submitted by

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is a record of Bonafide work carried out by them, in the partial fulfillment of the requirement for the award of Degree of Bachelor of Engineering (Computer Engineering) at Dhole Patil College of Engineering Pune under the Savitribai Phule Pune University , Pune. This work is done during the Academic year 2025-26.

Date:27/4/2026

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HOD

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ABSTRACT

This project focuses on designing and implementing a private Software as a Service (SaaS) cloud system over a Local Area Network (LAN) using Hadoop Distributed File System (HDFS). The main objective is to provide secure and efficient file storage and retrieval services within a controlled environment. The system allows users to upload and download files in a distributed manner, where large files are divided into smaller blocks and stored across the cloud system.

To enhance security, the files are encrypted before being stored in HDFS and decrypted during retrieval. A simple cloud controller is developed using open-source technologies to manage file operations and ensure proper distribution of data. This setup helps in improving data security, reliability, and accessibility within the local network.

The proposed system demonstrates how cloud computing concepts can be implemented on a small scale using existing LAN infrastructure, making it suitable for educational and laboratory environments.

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1. INTRODUCTION

Cloud computing is one of the most widely used technologies in modern computing systems, which provides on-demand access to computing resources such as storage, processing power, and applications over a network. Instead of storing data locally on individual systems, cloud computing allows users to store and access data remotely, making it more efficient, scalable, and flexible.

In this project, a private Software as a Service (SaaS) cloud is designed and implemented over a Local Area Network (LAN) in a laboratory environment using the Hadoop Distributed File System (HDFS). The main purpose of this system is to provide secure and efficient file storage and retrieval services within a controlled network environment. HDFS helps in storing large files by dividing them into smaller blocks and distributing them across multiple nodes, which improves reliability and fault tolerance.

This system demonstrates the practical implementation of cloud computing concepts on a small scale, making it useful for educational purposes. It helps students understand how cloud services work in real-world environments, including data distribution, security, and resource management.

1.1 Purpose of Private SaaS Cloud using HDFS:

The main purpose of this project is to design and implement a private Software as a Service (SaaS) cloud system over a Local Area Network (LAN) using Hadoop Distributed File System (HDFS). This system aims to provide a secure, efficient, and reliable platform for file storage and retrieval within a controlled environment such as a computer laboratory.

The project focuses on understanding and implementing basic cloud computing concepts like data distribution, virtualization of storage, and service-based architecture. By dividing files into smaller blocks and storing them in a distributed manner, the system improves data management, scalability, and fault tolerance.

Another important purpose of this project is to enhance data security by implementing encryption and decryption techniques during file upload and download operations. This ensures that sensitive data remains protected even within the network.

Overall, this project helps in gaining practical knowledge of cloud computing, HDFS architecture, and secure data handling in a private cloud environment.

1.1Overview of Evaluation system

This project presents the design and implementation of a private Software as a Service (SaaS) cloud system over a Local Area Network (LAN) using Hadoop Distributed File System (HDFS). The system is developed to provide a simple cloud environment where users can securely store and retrieve files within a controlled laboratory setup.

In this system, a cloud controller is used to manage all file operations such as upload and download. When a user uploads a file, it is first encrypted for security purposes and then divided into smaller blocks. These blocks are stored across the HDFS-based storage system in a distributed manner. This improves storage efficiency, reliability, and fault tolerance.

When a user requests to download a file, the system retrieves all related blocks from HDFS, reassembles them into the original file, and then decrypts it before providing access to the user. The entire process is handled over a LAN environment, making it suitable for institutional or lab-level cloud computing practice.

This project helps in understanding the working of cloud computing concepts such as SaaS, distributed storage, data security, and file management in a practical and simplified way using open-source technologies.

1.2 What is Private Saas Cloud Using HDFS

A **Private SaaS Cloud using HDFS** is a cloud computing system that is built for a **specific organization or lab (private use)** where software services are provided to users over a network, and data is stored using the **Hadoop Distributed File System (HDFS)**.

In this system:

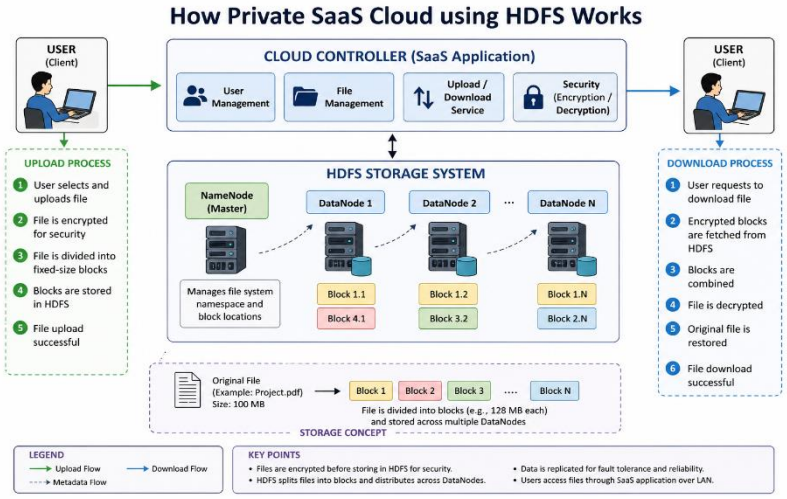
- **Private Cloud:** The cloud is restricted to a single organization or LAN environment, not open to the public.
- **SaaS (Software as a Service):** Users access software services (like file upload/download) through a web or client interface without managing the backend system.
- **HDFS (Hadoop Distributed File System):** It is used to store large files by breaking them into smaller blocks and distributing them across multiple nodes for reliability and efficiency

1.2.2 Importance of Private Saas Cloud Using HDFS

The importance of a Private SaaS Cloud using HDFS lies in providing a secure, efficient, and scalable way to store and manage data within a controlled environment such as a laboratory or organization.

- It helps in understanding real-time cloud computing concepts like SaaS and distributed storage.
- It provides secure data storage by using encryption during file upload and download.
- It improves data reliability and fault tolerance using HDFS, where files are divided into blocks and stored across multiple nodes.
- It enables easy file access and management over a LAN without depending on external cloud services.
- It is useful for educational purposes, helping students gain practical knowledge of cloud architecture and big data storage systems.
  Overall, this system demonstrates how modern cloud technologies can be implemented on a small scale for learning and practical use.

1.3 How the Evaluation Process Works:



2.Objectives Of Private Saas Cloud Using HDFS:

1. To design and implement a private SaaS cloud system over a LAN environment.
2. To develop a cloud controller for handling file upload and download operations.
3. To divide large files into smaller blocks for efficient storage and processing.
4. To provide secure data storage by implementing encryption and decryption techniques.
5. To ensure reliable and fault-tolerant file storage using distributed architecture.
6. To understand practical implementation of cloud computing concepts in a lab environment.
7. To improve knowledge of SaaS, HDFS, and cloud-based data management systems.

3. 🖥️ VMware ESXi Cloud Installation (Host/Node Setup)

In this project, we are using VMware ESXi to build and manage the private cloud infrastructure. ESXi is a hypervisor that allows multiple virtual machines to run on a single physical host system efficiently.

⚙️ ESXi Host/Node Installation Requirements

To install and run VMware ESXi successfully, the following hardware requirements must be satisfied:

🔹 1. Processor Requirements

- ESXi 6.7 requires a host machine with **at least two CPU cores**
- Supports **64-bit x86 processors**
- CPU must support **hardware virtualization**:
    - Intel VT-x or AMD RVI (SVM)
- Virtualization support must be **enabled in BIOS settings** .

🔹 2. Memory (RAM) Requirements

- Minimum required RAM: **4 GB**
- Recommended RAM for production environments: **8 GB or more**

- Higher RAM improves performance of virtual machines

3. Storage Requirements

- Requires a **SCSI disk or local storage** .
- Can also use **non-network RAID LUN** .
- Must have **unpartitioned free space** for installing ESXi and creating virtual machines .

4. Network Requirements

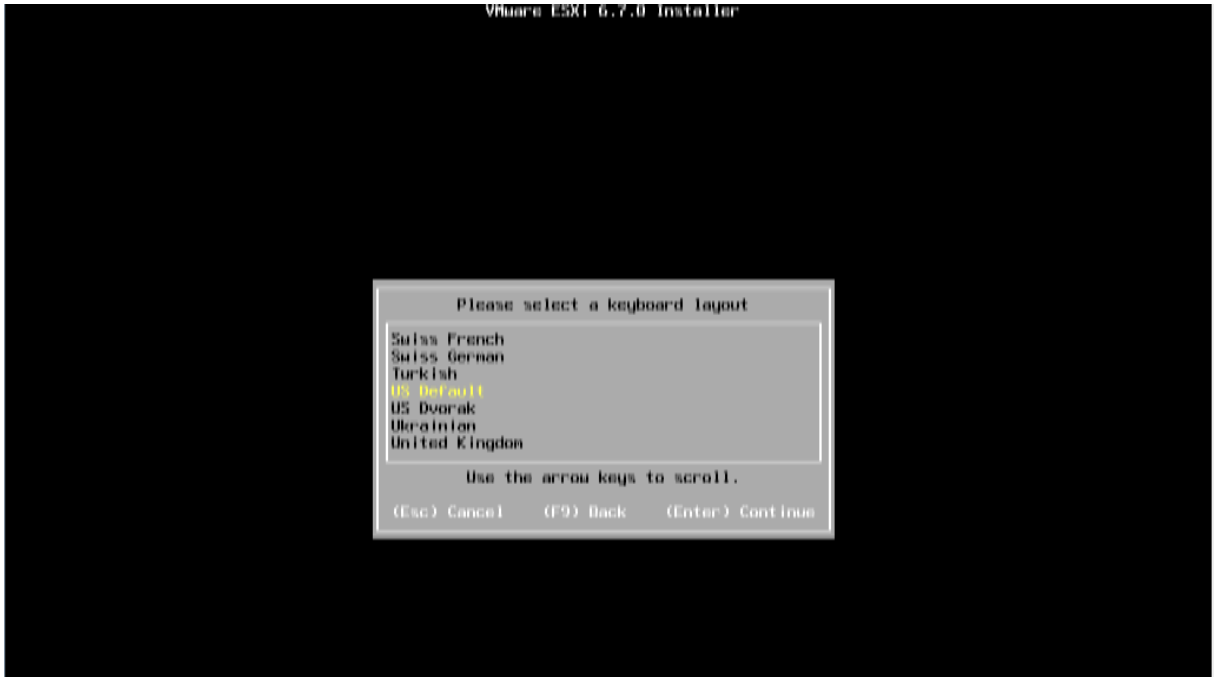
- One or more **Gigabit Ethernet (or faster) network adapters.**
- Supported network adapter models should be checked from VMware compatibility list .
- Required for communication between host machines and virtual machines .

ESXiInstaller: Accept

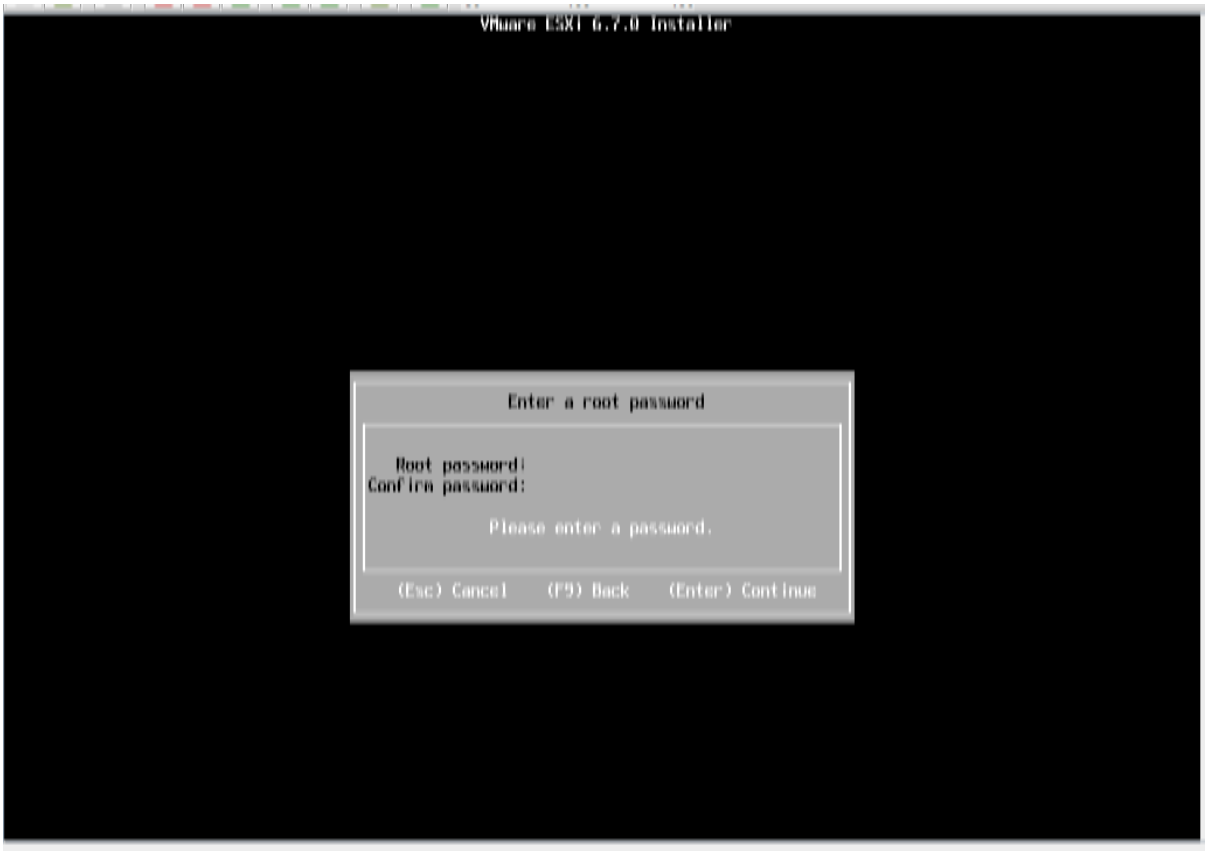
Agreement:



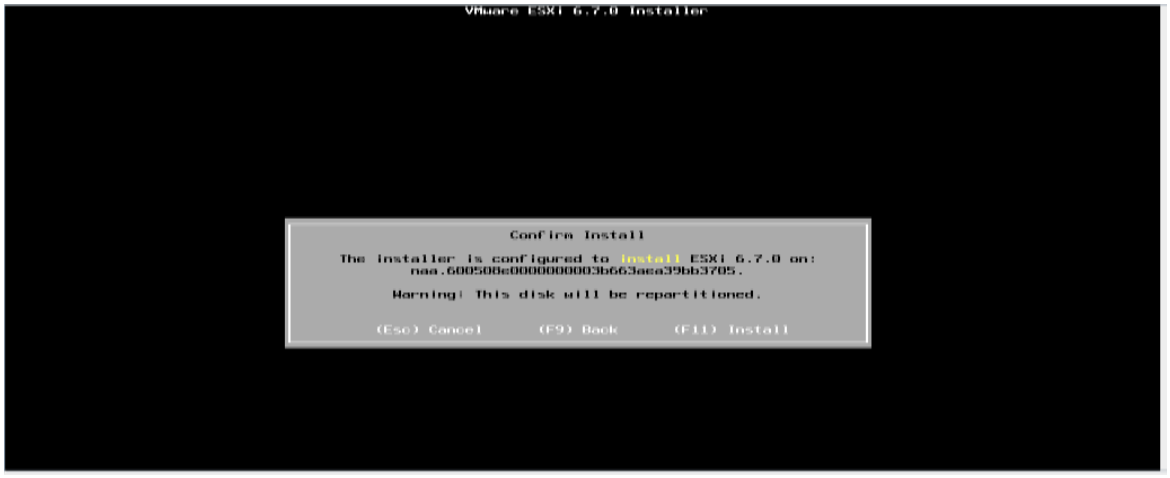
Select storage:



Keyboard Layout:



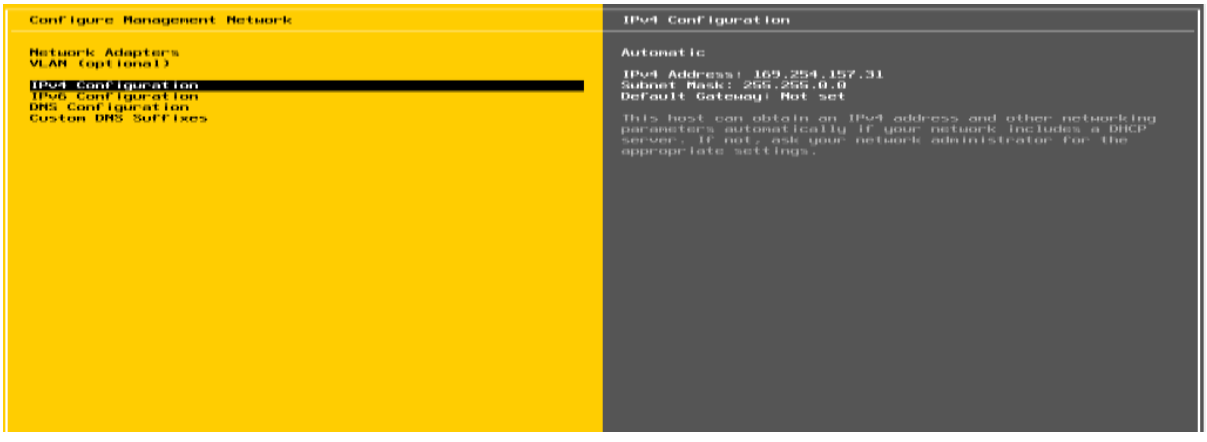
Set NodeESXi Root Password :



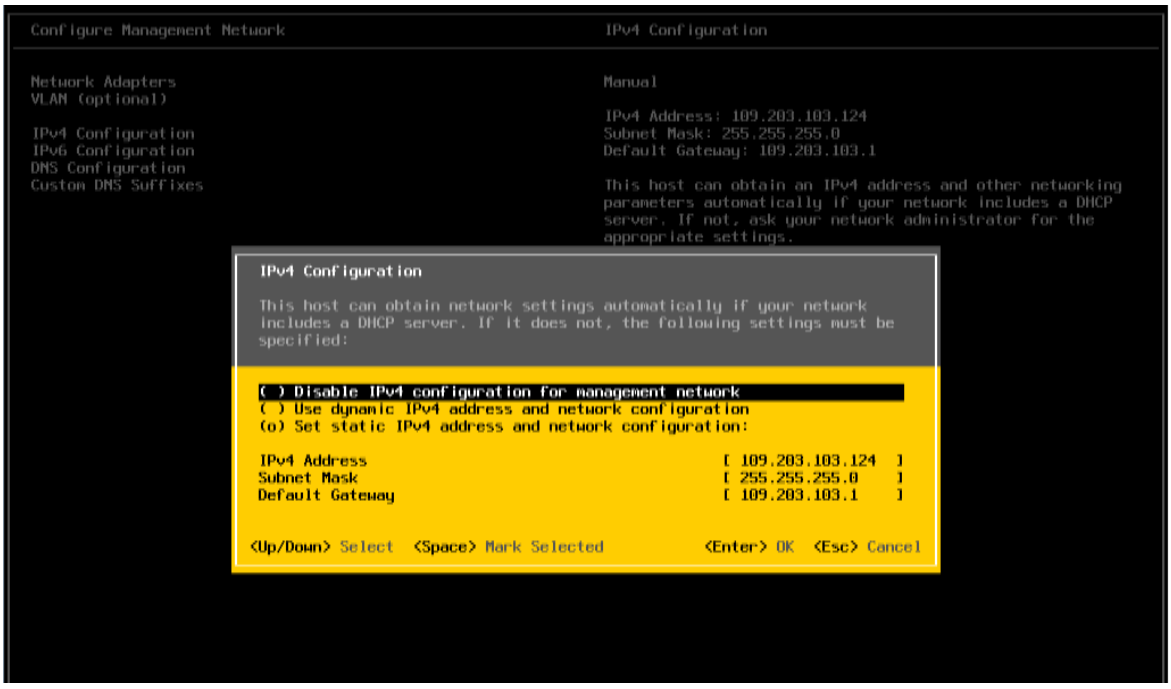
Installation complete (Reboot)CLII interface to configuration



CLI Interface to Configuration:



Configure Management Network

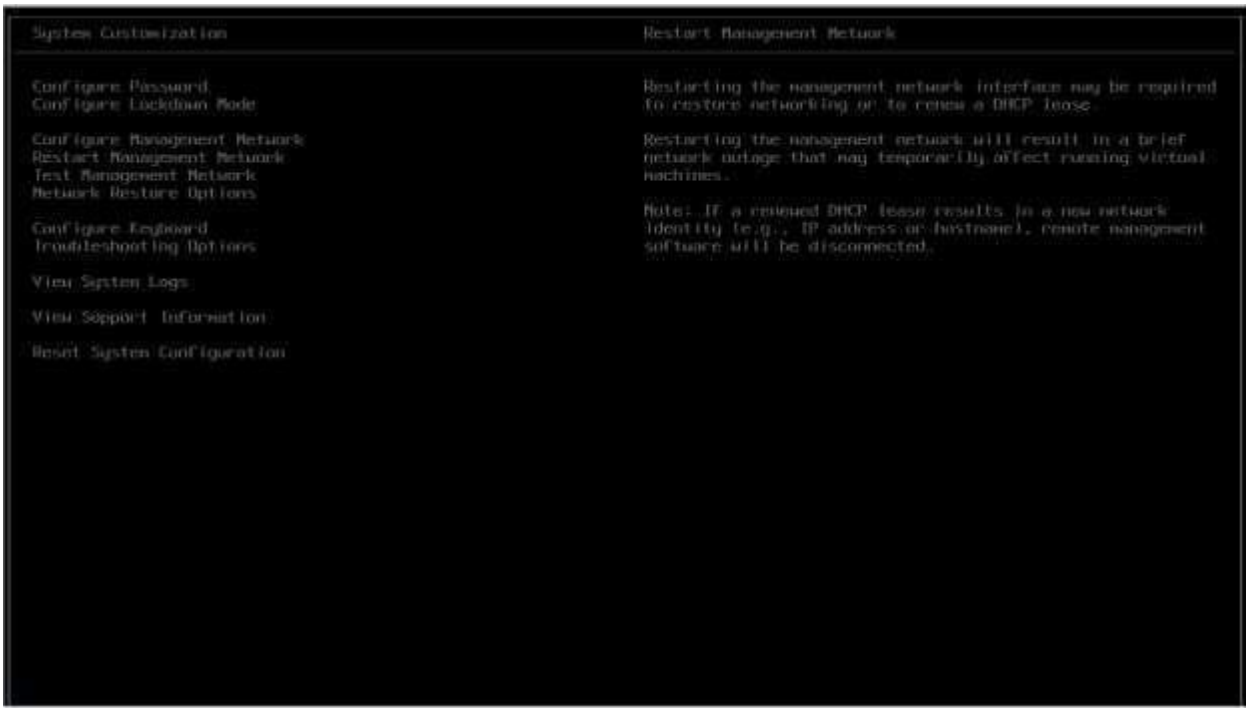


Set IPV4



Set DNSeriver :

Restart Management Network



GUIAccess :

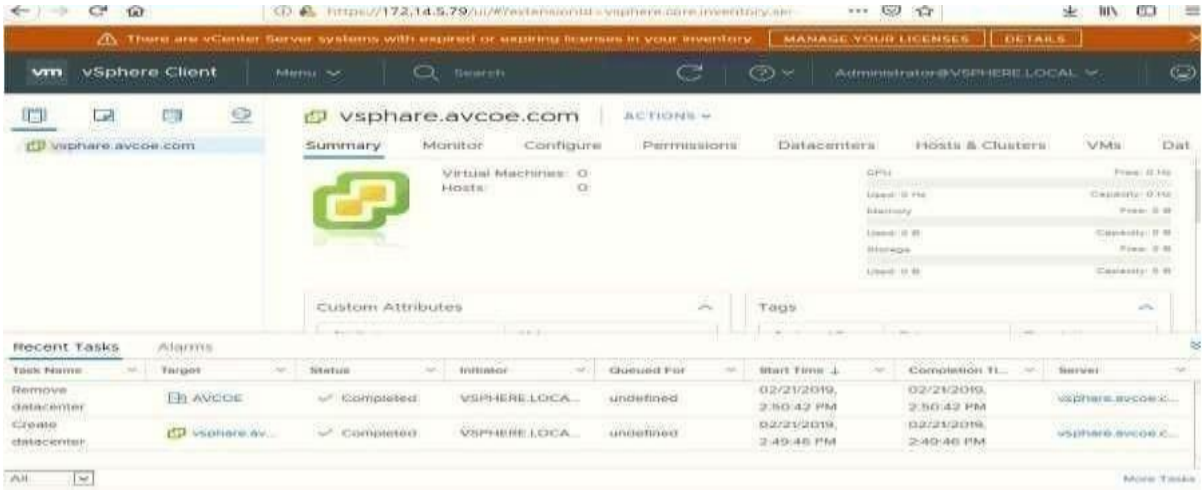


ClusterSetup

- CreatingDatacenter
- CreatingCluster
- Adding Hosts incluster

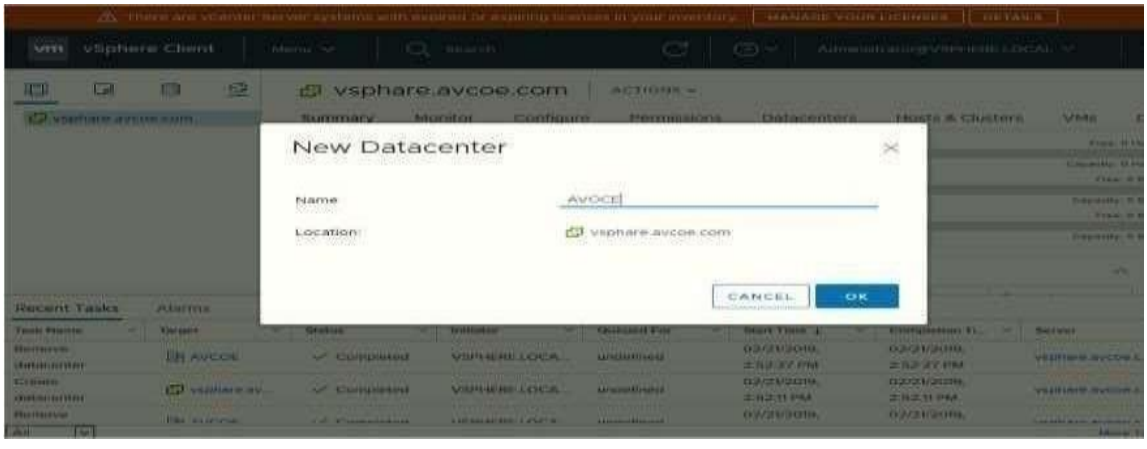
- Resourcesafteraddingcluster.
- DRS
- Failover

VCenter Access:



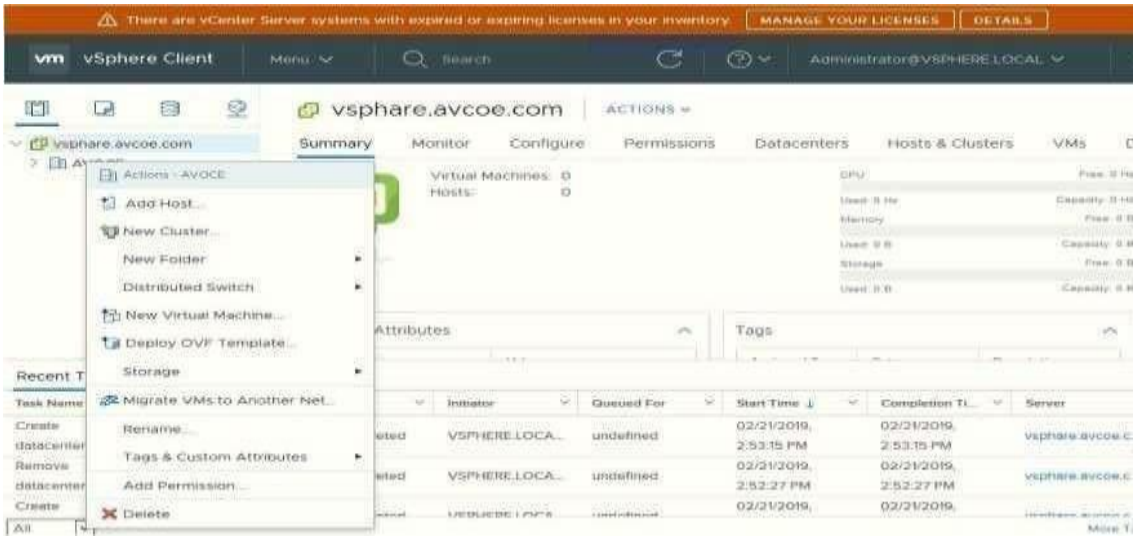
Create Datacenter:





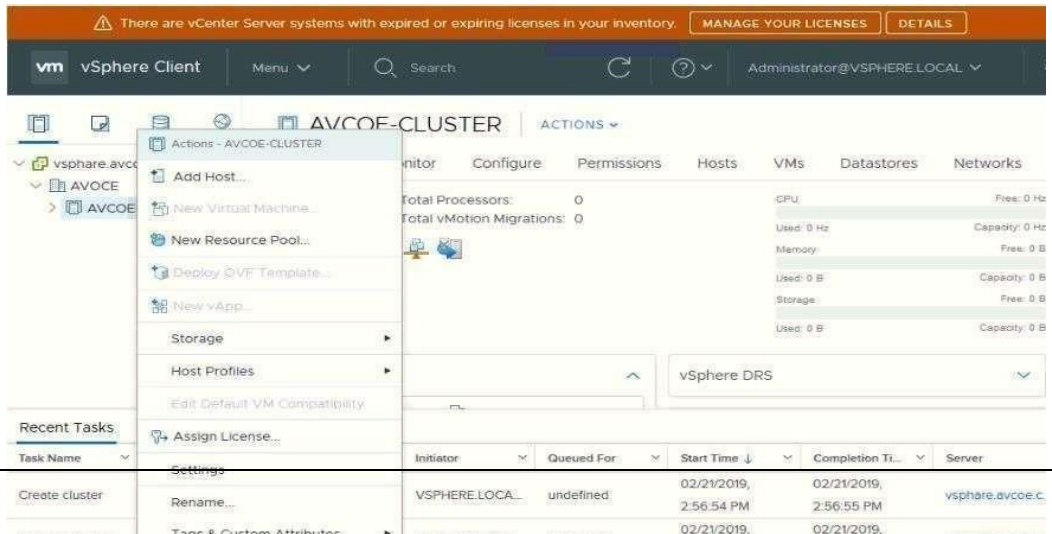
Create cluster :

Assign Cluster





Add Host



Add host IP :

Fi



Enter Host Credentials

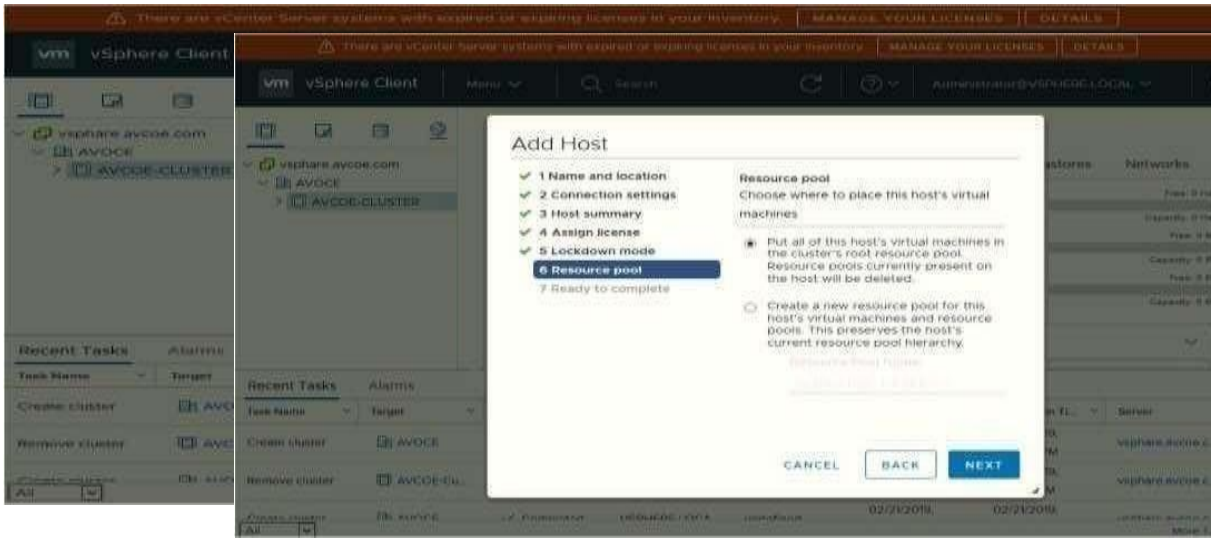


Hot Summary:

Lock Down

Mode:

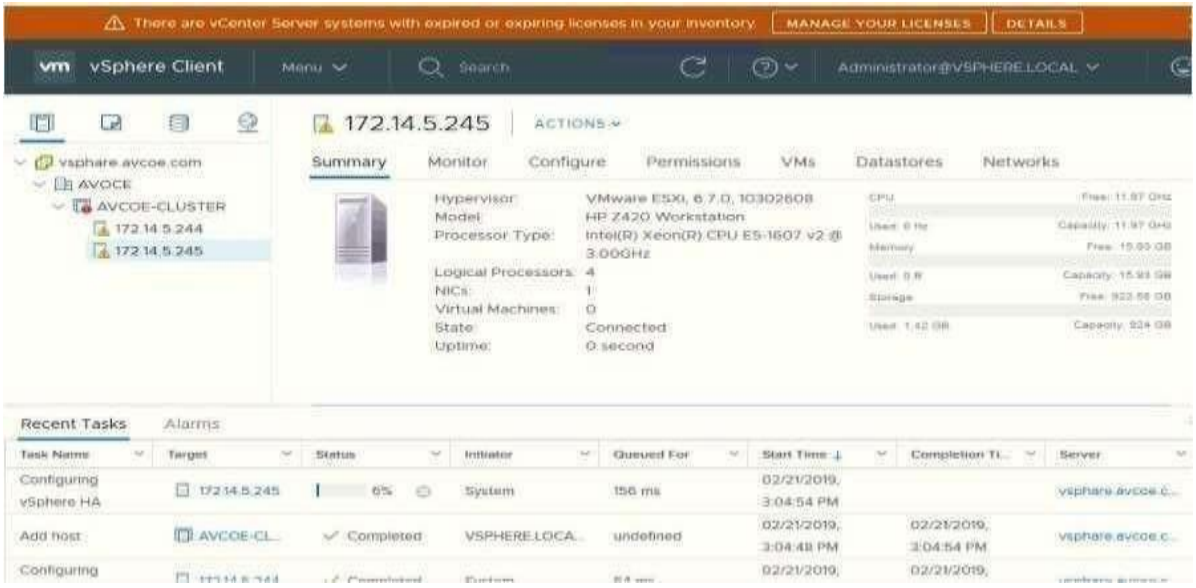
Add Host in Pool

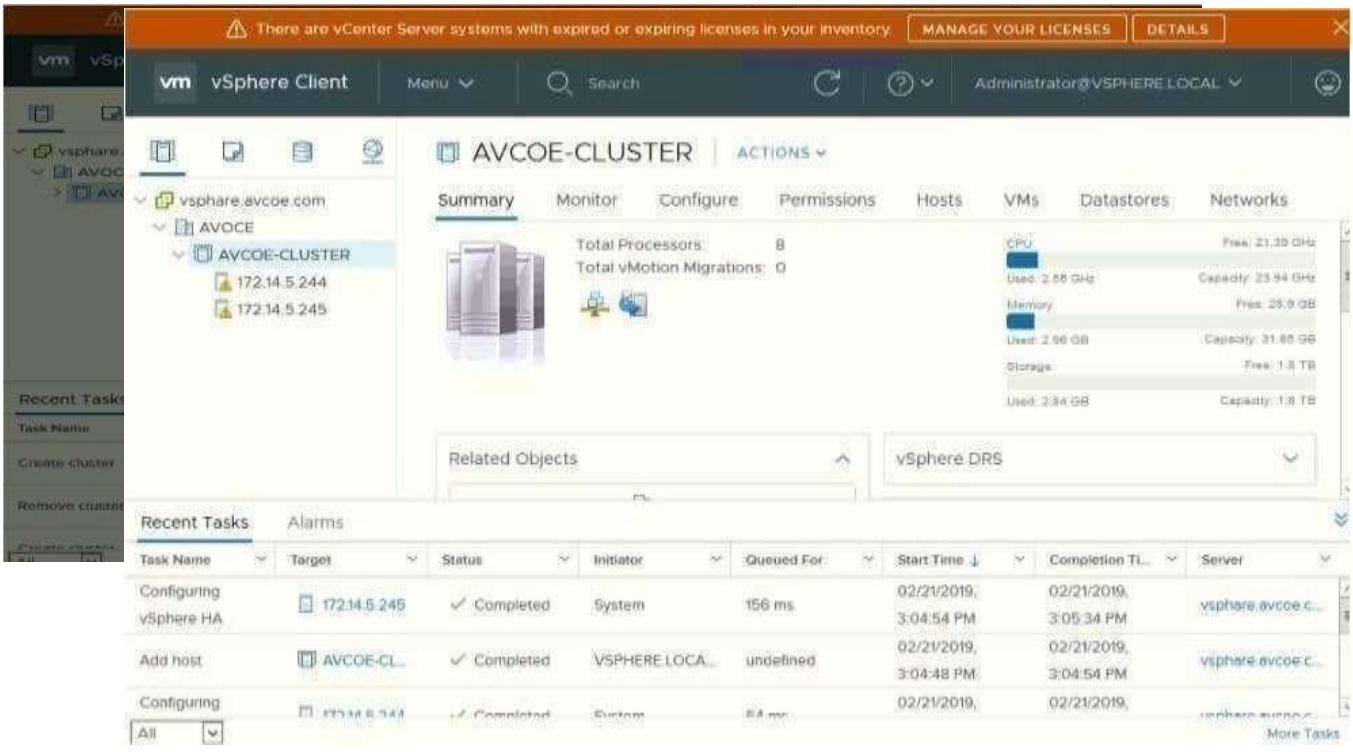


Finish

Host View and View Config:

Cluster View and Configuration:





6.Advantages & Disadvantages

☆ Advantages of Private SaaS Cloud using HDFS (with VMware ESXi)

- 🔒 High Security – Data is stored within a private network, reducing risk of external attacks.
- ⚡ Better Performance – Fast data access due to LAN-based infrastructure.
- 📦 Scalable Storage – HDFS allows easy expansion by adding new nodes.
- 💰 Cost Effective – Reduces dependency on expensive public cloud services.
- 🖥 Centralized Management – Easy control of users, storage, and resources.
- 🔄 Fault Tolerance – Data replication in HDFS ensures reliability.

✕ Disadvantages

- 🖥 High Initial Setup Cost – Requires powerful hardware and infrastructure.
- 🖥💻 Complex Configuration – Setup of ESXi, HDFS, and networking is difficult for beginners.
- 🔧 Maintenance Required – Needs continuous monitoring and system administration.
- 🔑 Limited Access – Only available within LAN (not accessible globally unless configured).
- ⚠ Hardware Dependency – Performance depends heavily on physical system capacity.

7. Applications

A Private SaaS Cloud using HDFS and VMware ESXi is widely useful in environments where secure, scalable, and centralized data management is required within a local network. Below are the detailed applications:

🏫 1. Educational Institutions

- Used in colleges and universities for storing student assignments, notes, and projects
- Faculty can upload and share study materials with students
- Helps in conducting online lab submissions within campus network
- Provides centralized storage for academic resources

🏢 2. Small and Medium Enterprises (SMEs)

- Used for internal document management systems
- Employees can store, access, and share company files securely
- Reduces dependency on external cloud services like Google Drive or AWS
- Improves collaboration between departments

🖥️ 3. Computer Laboratories

- Provides shared storage for programming assignments and practical work
- Students can access virtual machines through VMware ESXi
- Easy management of software and lab resources
- Ensures uniform environment for all users

🏥 4. Healthcare Systems (Hospitals & Clinics)

- Secure storage of patient records and reports
- Easy access for doctors and medical staff within hospital network
- Helps maintain privacy of sensitive health data
- Supports fast retrieval of large medical files (X-rays, scans)

📊 5. Research and Development Centers

- Used to store large datasets for research work
- Supports distributed data processing using HDFS
- Helps researchers collaborate within secure environment
- Useful for AI, ML, and data analytics projects

8. Future Scope

☁ Integration with Hybrid Cloud systems (Private + Public Cloud) .

📱 Development of mobile-based SaaS access applications.

🤖 Use of AI for smart storage management and optimization.

🔒 Implementation of advanced security features like encryption & blockchain.

🌐 Expansion to multi-location cloud clusters over WAN.

⚙ Automation using DevOps tools (Docker, Kubernetes integration).

9.Conclusion

The Private SaaS Cloud using HDFS with VMware ESXi provides a secure, reliable, and scalable solution for data storage and application hosting within a LAN environment. By combining virtualization (ESXi) and distributed storage (HDFS), the system ensures efficient resource utilization, fault tolerance, and centralized management of data and services.

This approach reduces dependency on external cloud providers while offering better control, privacy, and performance for organizations. It is especially useful for educational institutions, laboratories, and enterprises that require secure internal data handling and easy access to shared resources.

Overall, the implementation of a private SaaS cloud demonstrates how modern virtualization and distributed computing technologies can be effectively integrated to build a cost-efficient and secure cloud infrastructure**.**

10. References

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2. Apache Hadoop Documentation – Hadoop Distributed File System (HDFS) Architecture and Usage https://hadoop.apache.org/docs.

3. Cloud Computing: Concepts, Technology & Architecture – Reference book for cloud computing models (SaaS, PaaS, IaaS).

4. Red Hat Enterprise Linux Documentation – Useful for virtualization and server configuration concepts.

5. Oracle VirtualBox Documentation – Reference for virtualization basics and comparison with ESXi.